

Lead AI/ML Engineer & Data Scientist | Production ML & GenAI | Distributed Systems | GCP · Azure · GPU Computing

Lead AI and ML engineer and data scientist with **10+ years** of experience, building production AI/ML/LLM systems. Leads end-to-end AI delivery, combining hands-on model development with technical leadership to turn business problems into scalable, high-throughput, low-latency production systems. Experience spans predictive modelling, forecasting, operations research, computer vision, and generative AI across retail, cybersecurity, manufacturing, insurance, and supply-chain optimization.

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## Skills

- **LLM Training & Serving:** LLM Fine-tuning; LoRA / QLoRA; Evaluation; OpenAI-Compatible Serving
- **Modelling & Analytical Methods:** Classical ML; Deep Learning; Computer Vision; NLP; Forecasting; Feature Engineering; Operations Research; Autoencoders
- **Frameworks & APIs:** scikit-learn; XGBoost; PyTorch; TensorFlow; Hugging Face / PEFT; Unsloth; vLLM; OR-Tools
- **GPU & Distributed Computing:** Multi-GPU Training; RAPIDS cuDF / CuPy
- **Model Lifecycle:** MLflow; Evaluation / Registry; Retraining; Vertex AI / Azure ML Pipelines; Feature Store
- **Cloud & Deployment:** GCP / Azure; GKE / Kubernetes; Docker; Cloud Run / Functions; Azure Functions / Logic Apps; Helm / Istio
- **Reliability & Reporting:** CI/CD; Cloud Build; ArgoCD / GitOps; Prometheus / Grafana; Monitoring / Load Testing; Streamlit / Power BI
- **Languages, Data & Storage:** Python; SQL; Beam / Dataflow; Pub/Sub; BigQuery; Redis; GCS / Azure Blob; Pandas / NumPy; Dask; Git / Linux

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## Experience

### Lead Artificial Intelligence Engineer

Best Buy | Jul 2025 - Present

- Engineered a reusable **LLM fine-tuning framework** in Python with Unsloth, PEFT / Hugging Face Transformers, Accelerate and LoRA / QLoRA for distributed training, task-specific adapters and automated evaluation.
- Standardized task configuration and model artifacts so multiple tasks share one serving deployment.
- Deployed **vLLM serving on GKE**, with **estimated annual API-cost savings of ~\$300K**. Kept inputs company-managed, reducing dependence on external API availability and behavior during peak demand.
- Delivered OpenAI-compatible endpoints with Docker, Kubernetes, HPA autoscaling and Istio; monitored services with Prometheus and Grafana.
- Built **batch, streaming and real-time fraud systems** with a feature pipeline handling **40,000 events/sec at peak**, using Python / Java Apache Beam, GCP Dataflow, Pub/Sub and Vertex AI Feature Store.
- Supported a **200 ms scoring SLA** and improved holiday fraud-score coverage by **~20%**; deployed XGBoost through Helm and Seldon Core with JWT-secured endpoints.
- Led retail ML/LLM architecture and delivery with data scientists. Automated GitOps with Cloud Build, ArgoCD and Google KCC; managed artifacts in GCS and Artifact Registry.

### Lead ML Engineer - ML Engineering & Operations

Wipro | Aug 2023 - Jul 2025

- Engineered **AWS account digital fingerprinting** on NVIDIA DGX with RAPIDS cuDF / CuPy. Trained per-entity behavioral autoencoders on **billion-row security datasets**, with continuous retraining, MLflow model management and Grafana anomaly alerts.
- Scaled **Proofpoint phishing detection to 2,400 emails/sec (2x throughput)** through GPU-accelerated preprocessing, feature engineering and inference. Integrated analyst feedback and Microsoft Sentinel to detect missed phishing and support quarantine.
- Automated **abuse-inbox triage, doubling case volume** and reducing manual review. Used ML routing and Python OCR (pytesseract, cuDF) across email bodies and attachments; routed scammer identifiers to security teams and partners for takedowns.
- Mentored ML/MLOps engineers, coordinated client/NVIDIA delivery, screened **300+ resumes** and conducted **50+ technical interviews**. Earned Best Buy client AML Hall of Fame recognition (2024).

### Senior Data Scientist - Data Science / Azure ML

Accenture | May 2021 - Sep 2023

- Developed **BamaGruppen product-quality prediction** in Python on Azure ML, with automated retraining, Azure Blob storage, Logic Apps and stakeholder alerts.
- Increased truck inspections from **130 to 160/day**, defect detection by **17%** and customer approval by **20%**, delivering **\$2.5M annual savings**.
- Replaced manual Excel planning with automated last-mile routing and transport optimization using OR-Tools and mixed-integer models, integrating Azure Functions, Logic Apps and RPA to enable rapid replanning and improve delivery efficiency.
- Modelled **supply-chain and distribution-hub scenarios** across cost, freshness, carbon and warehouse volumes using Python / OR-Tools on Azure ML, Azure Blob and Power BI.
- Led cross-functional delivery, mentored data scientists and presented network-design recommendations to Norwegian stakeholders.

### Data Scientist

GD Research Centre Pvt Ltd | May 2020 - May 2021

- Developed **GlobalData forecasts** and scenario analysis across **15+ industry verticals** with Python forecasting and feature-engineering pipelines, informing market intelligence and resource allocation.

### Deep Learning Engineer

Tata Consultancy Services | Jul 2019 - Mar 2020

- Developed **semiconductor defect detection** with TensorFlow / Keras DenseNet on a GCP multi-GPU cluster. Distributed training on **~1 million images in ~3 hours** achieved **90% classification accuracy**.
- Ingested SQL Server data and served inference with Docker / TensorFlow Serving. Deployed the end-to-end system at Amkor, South Korea, and trained the client team.

### Senior Software Engineer - ML

Infosys | May 2016 - Jul 2019

- Implemented **opportunity win/lose prediction** with logistic-regression scoring and data pipelines for sales prioritization; integrated SQL Server Integration Services, SSAS and Power BI dashboards.
- Delivered tested **insurance ID-card workflows** using COBOL, JCL, DB2, VSAM and CICS, from impact analysis and design through implementation, support and team knowledge sharing.

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### Selected Recognition

Best Buy Talent Quotient (2026) - cybersecurity and fraud delivery.

Best Buy client AML Hall of Fame (2024), during the Wipro engagement - DGX framework delivery and offshore mentoring.

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### Certifications

- Google Cloud Certified: Generative AI Leader | Microsoft Certified: Azure AI Engineer Associate
- Microsoft Certified: Azure Data Scientist Associate | Microsoft Certified: Power BI Data Analyst Associate
- Oracle Cloud Infrastructure 2024 Generative AI Certified Professional

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### Education

B.E. Electrical & Electronics

Andhra University, Visakhapatnam | 2012 - 2016 · 80%